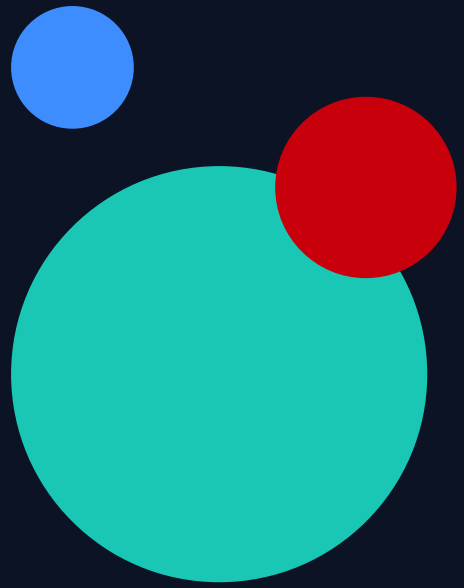
HUAWEI CLOUD • FINOPS

Intelligent capacity. Defensible cloud economics.

Executive report for an explainable cost, capacity and governance decision platform.



Murat Miraç Gedik • July 2026 • EN with TR summary

Executive scorecard

The deterministic demo identifies a material savings pool while preserving explicit capacity and governance controls.

$6.13K	$1.52K	24.79%	96.4
Current monthly cost	Estimated monthly savings	Savings opportunity	FinOps maturity score

DECISION IMPLICATION

A read-only pilot can validate the top recommendations, confirm capacity headroom and establish realized-savings evidence before any production change.

Source: reports/demo-analysis.json

The decision problem

BILLING IS LATE

Invoices and exports explain spend after consumption. They do not prove whether a workload was appropriately sized.

TELEMETRY IS PARTIAL

Utilization shows load but lacks ownership, price and budget context. A low average can hide a critical peak.

SAVINGS CAN OVERLAP

Multiple actions may target one resource. The platform reports only the highest modeled saving per resource.

The answer is one canonical decision record:

- cost + utilization + ownership + budget evidence
- risk and confidence attached to every recommendation
- human approval, rollback and post-change validation

Source: docs/optimization-methodology.md

A governed operating loop

1

OBSERVE

Normalize billing, inventory, utilization, tags and budgets.

2

DECIDE

Forecast, detect anomalies and rank explainable recommendations.

3

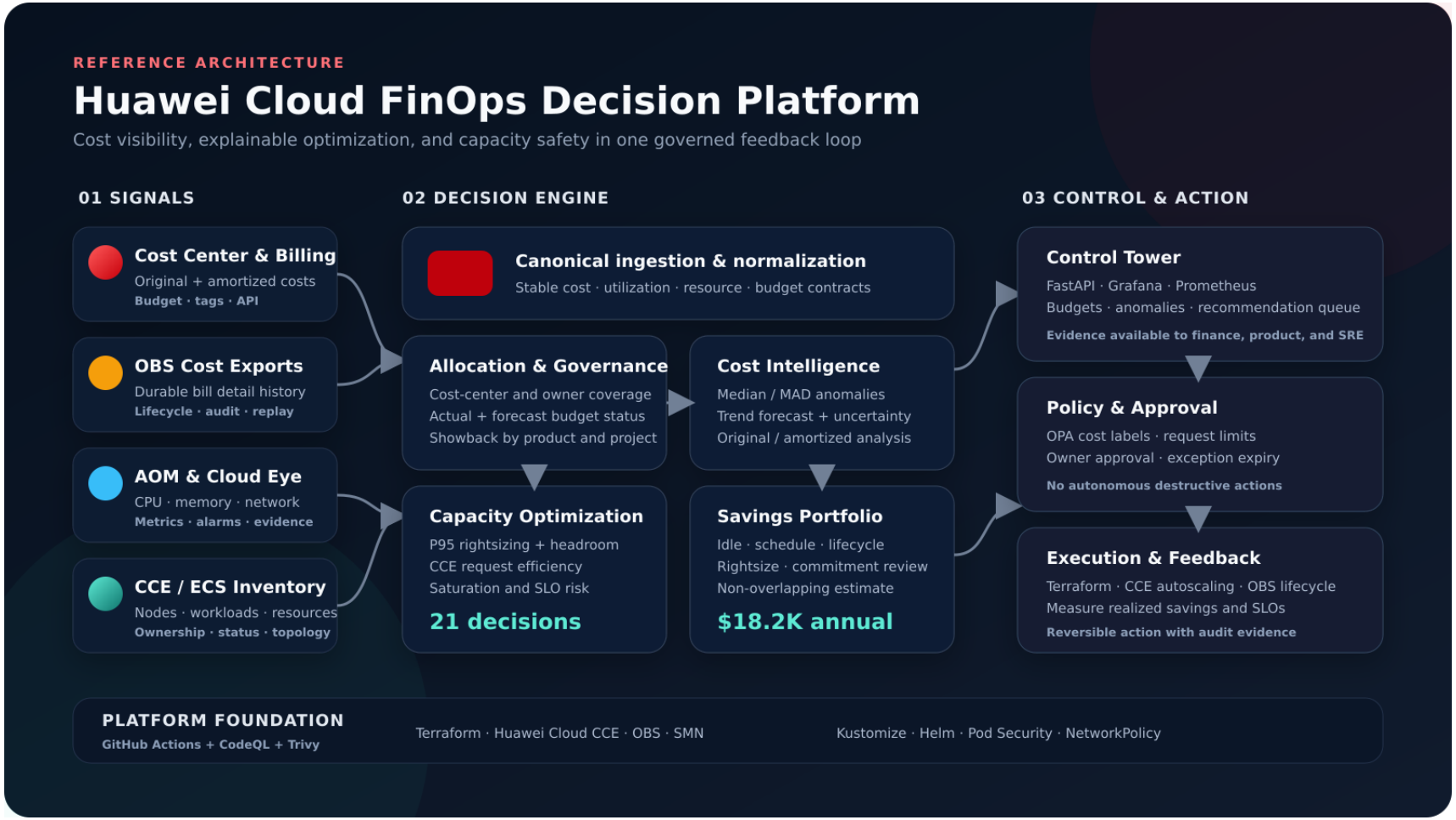
GOVERN

Assign owners, enforce policy, approve change and validate outcomes.

The analytics core is read-only. Execution remains an explicit, auditable operating decision.

Source: docs/finops-operating-model.md

Reference architecture



Source: docs/images/architecture.png; Huawei Cost Center: https://support.huaweicloud.com/intl/en-us/usermanual-cost/costcenter_0000001.html

Canonical data contracts

RESOURCE

Identity, service, region, owner, cost center, product, price mode and capacity.

COST

Actual and amortized daily cost, usage quantity, currency and resource link.

UTILIZATION

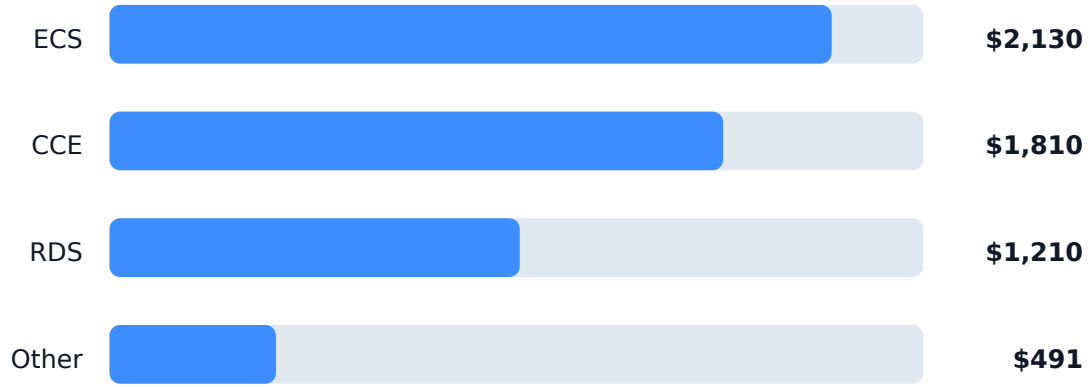
CPU, memory, network, disk and Kubernetes request saturation samples.

BUDGET

Scope, limit, actual threshold, forecast threshold and status.

Source: docs/data-contracts.md; Huawei Cost Analysis: https://support.huaweicloud.com/intl/en-us/usermanual-cost/costcenter_000002_09.html

Service economics



INTERPRETATION

ECS, CCE and RDS account for $5.15K of modeled monthly resource run-rate. Those same services contain $1.27K of conservative savings opportunity.

$5.15K

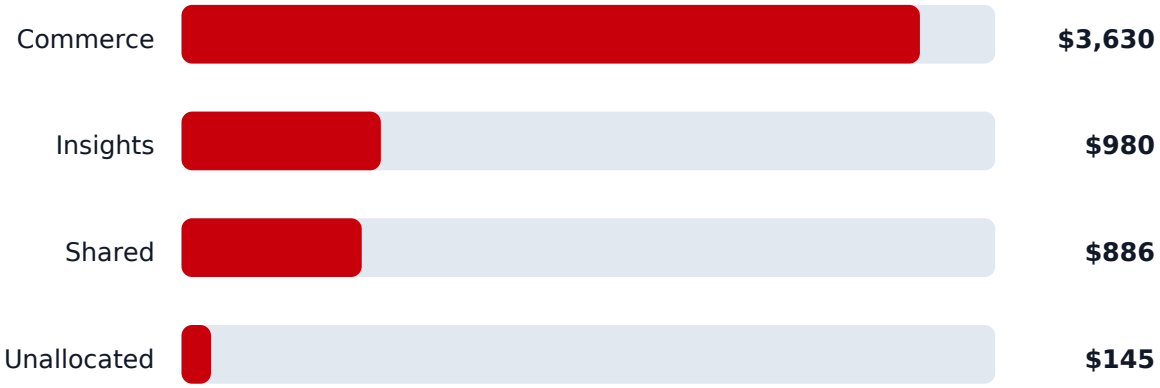
Top-3 spend

$1.27K

Top-3 savings

Source: reports/demo-analysis.json; data/demo/resources.csv

Allocation and ownership



97.43%

Cost allocation coverage

REMEDIATION

Missing cost-center, owner or product metadata creates tagging recommendations. Kubernetes admission policy prevents new unlabeled workloads.

Source: reports/demo-analysis.json; Huawei Cost Tags: https://support.huaweicloud.com/intl/en-us/usermanual-cost/costcenter_000005_01.html

Budget exposure

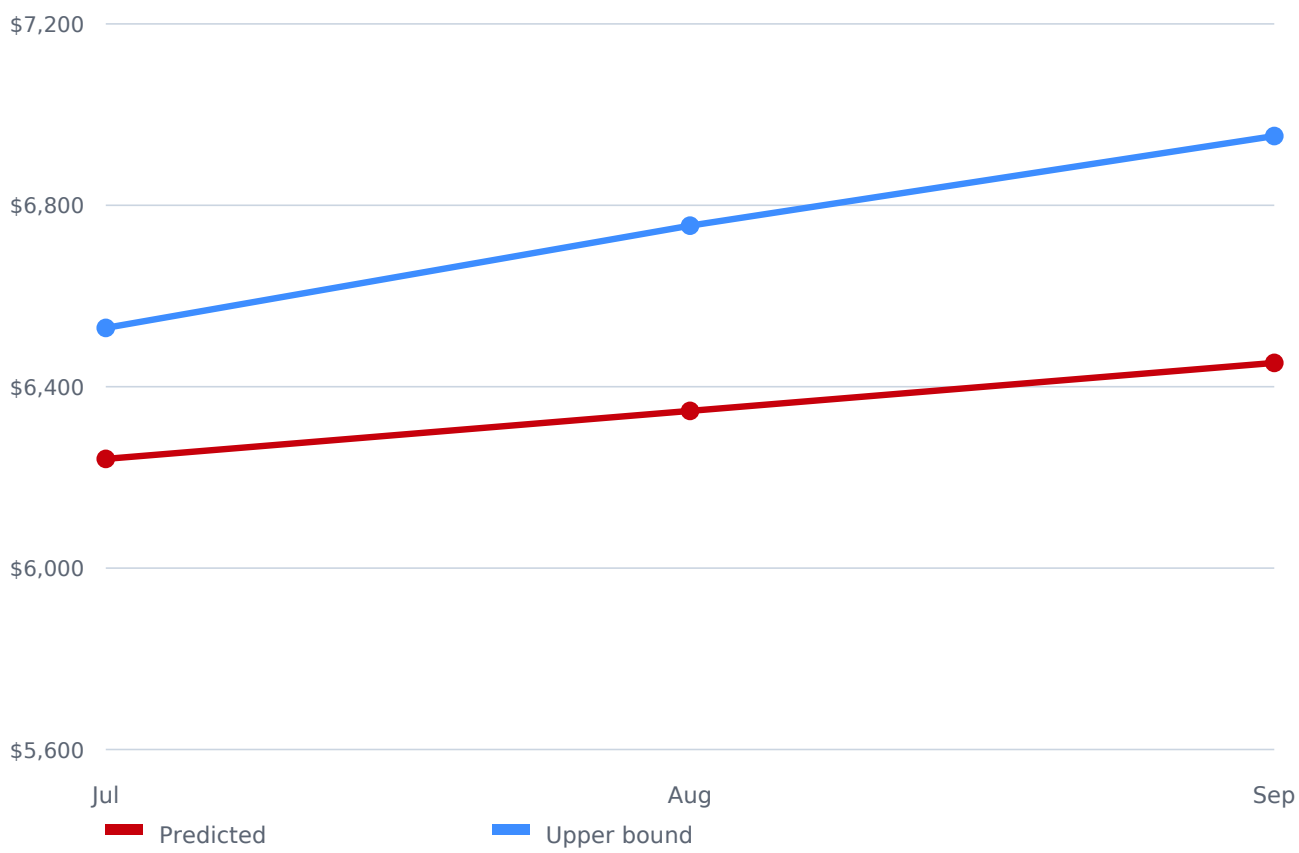
SCOPE	LIMIT	ACTUAL	FORECAST	STATUS
Non-Production	$550	133.66%	135.89%	BREACHED
Cloud Portfolio Monthly	$5,800	105.75%	107.60%	BREACHED
Commerce Product	$3,800	103.43%	105.49%	BREACHED
Insights Product	$1,350	79.94%	80.85%	HEALTHY

CONTROL RESPONSE

Prioritize non-production scheduling and idle cleanup; review Commerce rightsizing and commitment candidates; preserve the healthy Insights baseline.

Source: reports/demo-analysis.json; Huawei Budgets: https://support.huaweicloud.com/intl/en-us/usermanual-cost/costcenter_0000023_1.html

Three-month cost forecast



$6.24K

July point forecast

$6.95K

September upper band

Budget controls should use the uncertainty band, not only the point forecast.

Source: reports/demo-analysis.json; src/finops/services/forecast.py

Cost anomaly review

ML WORKLOAD

Date: 2026-06-21
Actual: $32.99
Expected: $17.05
Deviation: +93.55%
Severity: HIGH

CCE PRODUCTION NODE

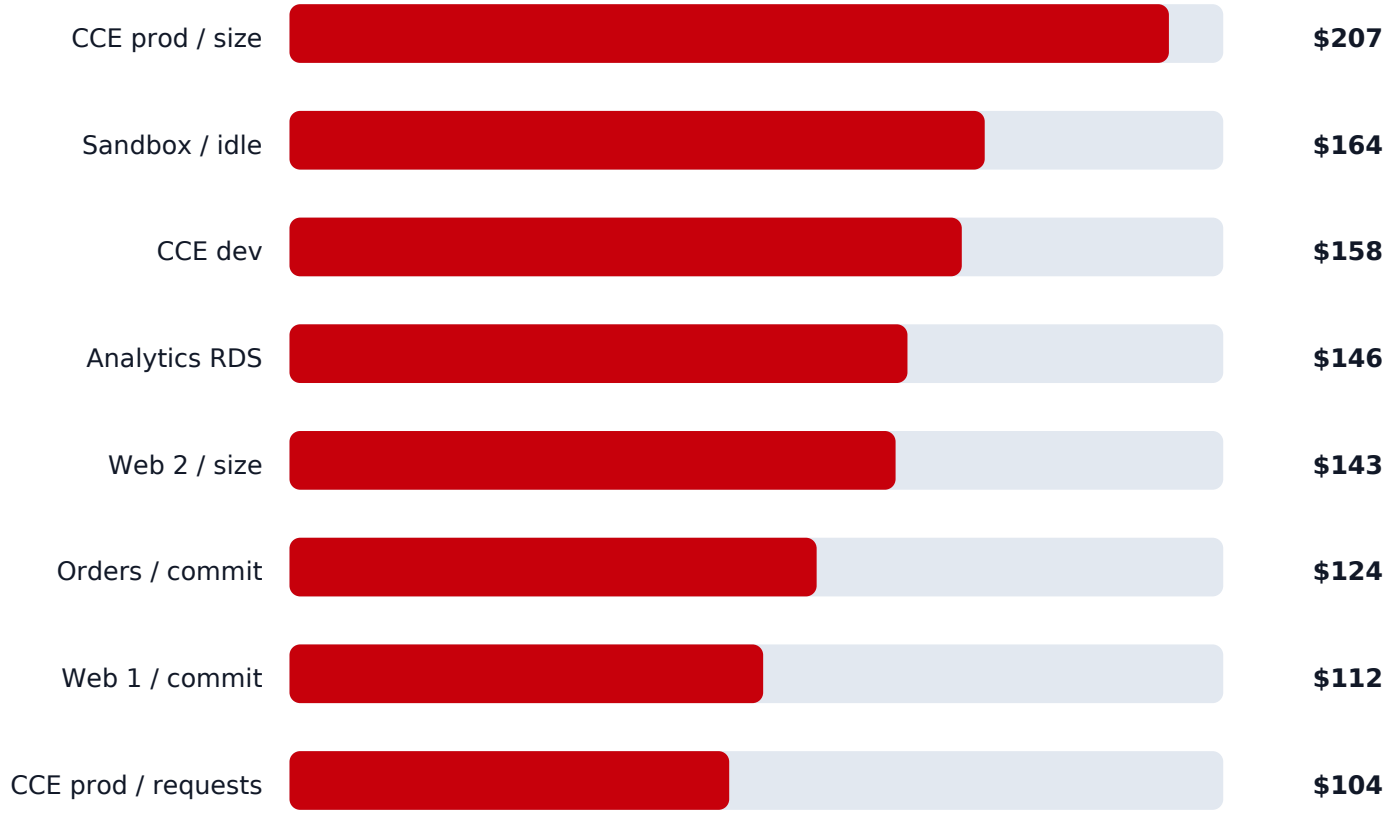
Date: 2026-05-18
Actual: $69.21
Expected: $25.48
Deviation: +171.68%
Severity: CRITICAL

An anomaly is an investigation trigger, not a root-cause claim.

- confirm owner, release or workload event
- compare usage and capacity telemetry
- document resolution and baseline effect

Source: reports/demo-analysis.json; docs/runbooks/cost-anomaly.md

Prioritized savings portfolio



$18.25K

Annualized opportunity

CONSERVATIVE TOTAL

The executive total selects the single largest modeled saving per resource. This prevents schedule, rightsizing and commitment actions from being added twice.

Source: reports/demo-analysis.json

Recommendation logic

RIGHTSIZE

Use P95 CPU, memory and required headroom; avoid mean-only decisions.

ELIMINATE WASTE

Release idle assets, schedule non-production and lifecycle object storage.

PRICE BETTER

Consider commitments only for stable, reviewed production demand.

PROTECT CAPACITY

Flag saturation and tune CCE requests before scaling down.

Source: src/finops/services/optimization.py; docs/optimization-methodology.md

Capacity guardrails

P95

Peak-aware utilization evidence

25%

Default capacity headroom

1 HIGH

Capacity-risk recommendation

ORDER OF OPERATIONS

1. Protect service health. 2. Validate request and limit efficiency. 3. Confirm autoscaling behavior. 4. Only then evaluate scale-down or commitment changes.

Source: reports/demo-analysis.json; Huawei CCE node pools: https://support.huaweicloud.com/intl/en-us/usermanual-cce/cce_10_0081.html

Security and control model

RISK	CONTROL	RESIDUAL DECISION
Credential leakage	Runtime secret or agency; no secrets in repo	Rotate and audit
Destructive action	Read-only engine; human approval	Named change owner
Runaway scaling	Requests, limits and HPA ceiling	Capacity review
Unallocated cost	Required owner and cost-center labels	Tag remediation
Supply chain	CodeQL, Trivy and dependency updates	Review findings

NON-GOAL

The platform does not delete, resize or purchase cloud resources automatically.

Source: docs/security-threat-model.md; policies/finops/kubernetes.rego

Huawei Cloud deployment model

COST & DATA

Cost Center export or API adapter
OBS retention and lifecycle
Canonical analysis data

RUNTIME

SWR container image
CCE deployment, HPA and PDB
Network policy and non-root pod

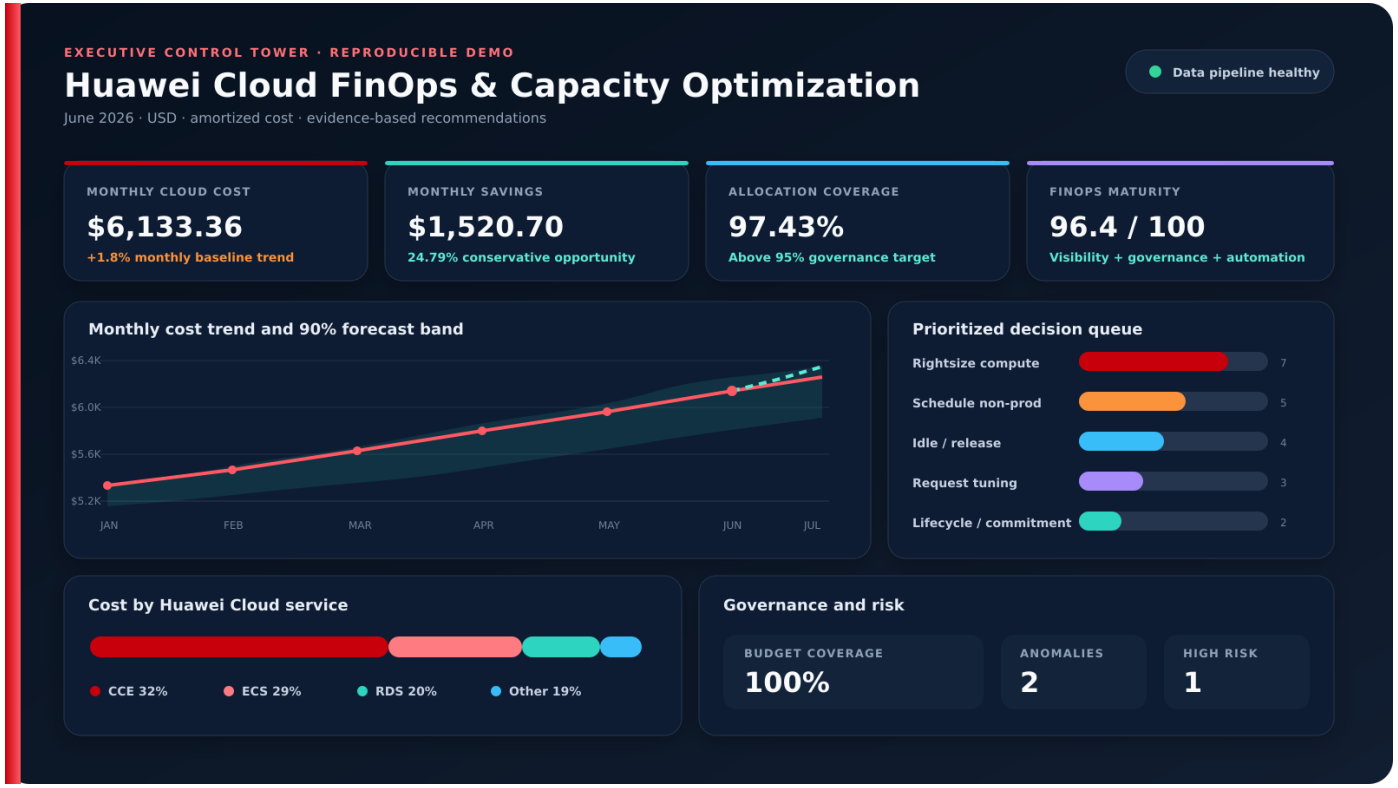
OPERATIONS

Prometheus-compatible metrics
Grafana dashboard and alerts
SMN notification integration

Terraform, Helm and Kustomize provide reproducible deployment paths without embedding credentials.

Source: docs/deployment-huawei-cloud.md; infra/terraform

FinOps observability



DECISION SIGNALS

Current cost
Savings opportunity
Allocation coverage
Budget status
Anomaly count
Capacity risk

OPERATIONS

Health probes, Prometheus metrics and alert rules keep the FinOps service observable.

Source: docs/images/dashboard-preview.png; observability/

Engineering validation

CONTROL	SCOPE	RESULT	GATE
Unit tests	53 scenarios	PASS	required
Coverage	branch-aware	97.09%	>= 90%
Repository	JSON/YAML/demo	PASS	required
API	health + analysis	PASS	required
Terraform	fmt + validate	CI	required
Container	build + smoke	CI	required
Security	CodeQL + Trivy	CI	required

LOCAL EVIDENCE VS CI

Python tests, coverage and repository contracts are verified locally. Terraform, container and security workflows execute after publication.

Source: docs/validation.md; .github/workflows/

30-day read-only pilot

WEEK 1

Confirm sources, owners and allocation baseline.

WEEK 2

Review top 10 recommendations and capacity evidence.

WEEK 3

Approve low-risk quick wins and document rollback.

WEEK 4

Measure realized savings and operating impact.

Success criteria

- 100% of top recommendations have owners
- realized savings are separated from modeled savings
- no action reduces required service headroom
- Finance, Platform and product teams accept the evidence log

Source: docs/finops-operating-model.md; ROADMAP.md

Türkçe yönetici özeti / Sources & scope

TÜRKÇE ÖZET

- Demo portföy aylık 6.133,36 USD maliyet ve 1.520,70 USD tasarruf fırsatı üretmektedir.
- Toplam fırsat oranı %24,79; yıllıklandırılmış potansiyel 18.248,40 USD'dir.
- Üç bütçe kapsamı aşılmış, iki maliyet anomalisi ve bir yüksek kapasite riski belirlenmiştir.
- Karar motoru salt-okunurdur; değişiklikler sahip, onay, geri dönüş ve doğrulama gerektirir.
- Önerilen sonraki adım 30 günlük, üretimde otomatik değişiklik yapmayan pilot çalışmadır.

PRIMARY SOURCES

- support.huaweicloud.com/.../costcenter_0000001.html (Cost Center)
- support.huaweicloud.com/.../costcenter_000002_09.html (Cost Analysis)
- support.huaweicloud.com/.../costcenter_000005_01.html (Cost Tags)
- support.huaweicloud.com/.../costcenter_0000023_1.html (Budgets)
- support.huaweicloud.com/.../cce_10_0877.html (CCE Cost Governance)
- support.huaweicloud.com/.../cce_10_0081.html (CCE Node Pools)
- reports/demo-analysis.json (deterministic project evidence)

SCOPE NOTE

All financial figures are synthetic demo outputs and modeled opportunities. They are not realized customer savings or production performance claims.

Source: Repository documentation and official Huawei Cloud references